Derivation for time period of simple pendulum using Dimensional Analysis:

There are three factors on which the time period of a simple pendulum depends:-

mass of the bob = m

Length of the string = L

acceleration due to gravity = g

Hence,

$$T \propto m^a L^b g^c \rightarrow (i)$$

$$[T] \propto [M]^a [L]^b [LT^{-2}]^c$$

$$[T] \propto [M]^a [L]^b [L]^c [T]^{-2c}$$

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$$[T] \propto [M]^{a} [L]^{b+c} [T]^{-2c}$$

equating powers we get

$1 = -2c$; $0 = b + c$; $\boxed{a = 0}$

$\boxed{c = -\frac{1}{2}}$; $0 = b - \frac{1}{2}$;

; $\boxed{b = \frac{1}{2}}$

putting values in eq (i)

$$T \propto m^{0} L^{1/2} g^{-1/2}$$

$$T \propto \left(\frac{L}{g}\right)^{1/2}$$

$$T \propto \sqrt{\frac{L}{g}} \Rightarrow T = K\sqrt{\frac{L}{g}}$$

where K is proportionality constant and its value is 2π

$$\boxed{T = 2\pi\sqrt{\frac{L}{g}}}$$